

Section 2

Emergent Space

Correlation Geometry

The third deliverable of Phase B: a fully merged, end-to-end rewrite of Section 2 against Construction Spec v1.0. Constructs **spatial geometry** from the GNS inner product via the **pairwise (Quadratic) kernel** $K_\omega(A,B)$ and the **irreducible (Cubic) kernel** $K_\omega(A,B,C)$; derives the **triangle inequality** from Cubic Volumetric Consistency (closing Open Target 1); builds the **metric quotient** on correlation-equivalence classes; reconstructs **observable-dependent direction**; selects **$D = 3$** via the closure defect $\Xi_C(D)$; and ports the **Type-Sign Coupling** and **Three-Layer Bridge C_e** from the Gen 3 amendment.

PHASE B - Section 2 Merge

Resolves the §0.8 forward-reference contract for K_ω ; certifies §1.3.5 and §1.7.3.

SCOPE · 9 Subsections · 4 Layers · 1 Quarantined Conjecture

SOURCE SPEC RCF-CONST-SPEC-v1.0, Ch. 5-9

DEF 2.1.1 RESTATED

SOE PHASE PROTECTION

THM 2.3.3 DERIVED

OPEN TARGET 1 CLOSED

§0.8 FWD-REF RESOLVED

$D=3$ CLOSURE DEFECT

Preamble — How to Read This Section

This document is the merged canonical form of Section 2 of the Reconciliation Causal Framework (RCF). It is the third deliverable of Phase B as specified in *RCF Unified Construction Specification v1.0*, and it builds directly on the closed foundations of *RCF Section 0 — Merged Canonical Form v1.0* and *RCF Section 1 — Causal Foundation v1.0*. Section 0 produced the kinematic algebra, the GNS representation, the Reconciliation Propagator $\mathbf{R}_t$, the thin physical sub-algebra $\mathbf{A}_{\text{phy}}^{\text{thin}}$ (§0.6), and the full physical sub-algebra $\mathbf{A}_{\text{phy}}$ (§0.7, certified equal to the thin candidate by Theorem 0.7.3). Section 1 then introduced the strict partial order of causal dependency $<$ (§1.1), the two-scale (SOE/MOE) speed limit (§1.3), the Open Extension Principle using $\delta\hat{\mathbf{C}}$ (§1.4), the Two-Link separation between causal and correlation links (§1.5), and the conditional emergence of Lorentzian structure (§1.7). Section 2 now constructs the complementary structure: emergent spatial geometry from the GNS inner product.

The structure follows the spec's source map (Table 4.1) row-by-row. Each subsection opens with a layer badge identifying its position in the $\mathbf{L} \rightarrow \mathbf{Q} \rightarrow \mathbf{C} \rightarrow \mathbf{Q}$ emergence ladder, a one-line source citation, and the epistemic tag inherited from the master manuscripts. Body text is ported verbatim where possible; rewritten passages are flagged inline with a spec chapter reference (e.g. *per Ch. 8*). The principal source for this merged section is **RCF_n.pdf §2.0–2.8** (the Gen 1 master manuscript, which uniquely contains the Cubic Volumetric Consistency theorem, the quotient-metric construction, and the closure-defect proof of $D=3$); this is augmented throughout by the post-amendment compact declaration **Section_2_2.pdf** for the SOE-protection of phase structure, the Type-Sign Coupling lemma, and the Three-Layer Bridge $\mathbf{C}_\varepsilon$ introduced by the Gen 3 amendment.

Dependency contract with Section 0

Section 0 v1.0 split $\mathbf{A}_{\text{phy}}$ into a thin candidate (§0.6, defined early by $\ker(\mathbf{M})$ -compatibility) and a full algebra (§0.7, defined late as the fixed-point of $\mathbf{R}_\infty$). **This merged Section 2 uses the thin candidate throughout:** Def 2.1.1 (localisable observables) is restated to require only $\mathbf{A}_{\text{phy}}^{\text{thin}}$ from §0.6, not the full $\mathbf{A}_{\text{phy}}$. This is consistent with the Section 1 restatement of Def 1.1.1 (zero-preserving event) and keeps the emergence chain acyclic: $\S 0.4 \rightarrow \S 0.5 \rightarrow \S 0.6 \text{ (thin)} \rightarrow \S 1.1.1 \rightarrow \S 2.1.1 \rightarrow \S 2.2 \rightarrow \S 2.3$. The phase structure of $\mathbf{K}_\omega$ (§2.1.4) is shown to be protected by the SOE spectral flow (§0.4.1) as a Layer B dynamical invariant, but the bare kernel definition does not require the full fixed-point algebra.

Forward-reference contract with §0.8 — FULLY RESOLVED

Section 0 v1.0 left a forward reference from §0.8 (Reconciliation Principle) to two Cubic ingredients: **(i)** the causal order $\prec$, and **(ii)** the correlation kernel K_ω . Section 1 supplied ingredient (i). **This merged Section 2 supplies ingredient (ii):** §2.1 defines K_ω rigorously on the localisable sector, §2.2 derives the exact emergent distance d_ω , and §2.3 proves the triangle inequality under Cubic Volumetric Consistency (closing Open Target 1, certifying the conditional theorems §1.3.5 and §1.7.3). With both ingredients in place, the variational target I(S) of §0.8 is fully grounded and the Reconciliation Principle becomes operational — the Quartic layer of the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder is now live.

Table of Contents

§2.0 Purpose of Correlation Geometry and Emergent Space	4
§2.1 The Correlation Kernels	7
§2.2 The Exact Emergent Metric	10
§2.3 Metricity Under Cubic Volumetric Consistency	12
§2.4 Emergent Points and the Metric Quotient	14
§2.5 Approximate Metricity under Coarse-Graining	16
§2.6 Observable-Dependent Direction	18
§2.7 Dimensional Closure: Relational Inference Selects Three Spatial Dimensions	20
§2.8 Type-Sign Coupling and the Three-Layer Bridge	23
§2.9 Architectural Summary	26

§2.0 Purpose of Correlation Geometry and Emergent Space

LAYER L → Q

Source: RCF_n.pdf §2.0 (Gen 1 master manuscript, integrated with Section_2_2.pdf §2.0 SOE/MOE split). Epistemic tag: [Established].

Section 1 developed the first post-foundational structures: the physical zero sector, zero-preserving observables, causal dependency, the Two-Link Principle, and open extension. These gave the framework a pre-geometric causal skeleton and an emergent speed limit. However, causal order alone is not space. A relation such as $e_i < e_j$ tells us that e_j depends on e_i ; it does not tell us how far apart two events are, and it does not define a metric, topology, dimension, locality, or spatial region. It also does not define distance between events that are not causally comparable. Thus, after causal dependency has been separated from relational correlation by the Two-Link Principle, the next task is to develop the correlation side.

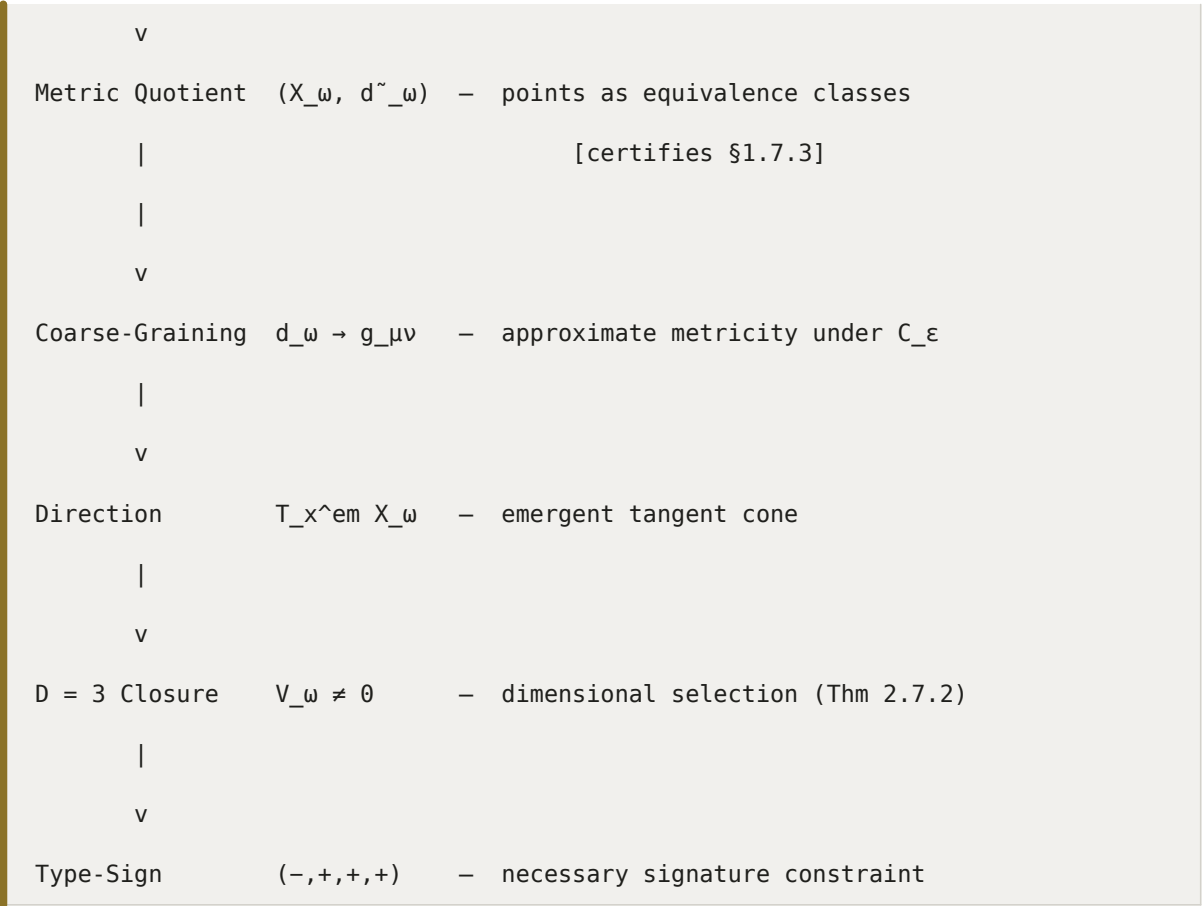
Space is not assumed as a background arena; it is reconstructed from correlation structure.

The framework does not begin with points in a manifold. It begins with admissible relational structure. What later appears as spatial distance is to be derived from how strongly physical observables, events, or localisable relational contexts are correlated in a physical state. This subsection establishes the conceptual chain by which the framework climbs from the Quadratic layer (pairwise correlation) through the Cubic layer (volumetric closure, metricity, direction, D=3 dimensional selection) to the threshold of the Quartic layer (Type-Sign Coupling and the Three-Layer Bridge C_ϵ that hands the geometry over to General Relativity in the coarse-grained limit).

§2.0.1 The Hierarchy of Geometric Emergence

The geometric emergence ladder of this section is strict and non-circular. Each rung depends only on the rungs below it and on the closed foundations of Sections 0 and 1. The rungs are:

The L → Q → C → Q' Geometric Emergence Ladder			
Quadratic Kernel	$K_\omega(A,B)$	–	pairwise existence, position
	v		
Cubic Kernel	$K_\omega(A,B,C)$	–	3-point direction, volume
	v		
Exact Metric	$d_\omega(A,B)$	–	logarithmic distance
	v		
Volumetric Closure		–	triangle inequality (Thm 2.3.1)
			[closes Open Target 1]



Each arrow in this ladder is a one-way derivation. The Cubic Kernel is constructed from the Quadratic Kernel by adjoining a third link and evaluating the GNS triple-product. The Exact Metric is defined as the negative logarithm of the Quadratic Kernel. The triangle inequality is derived (not assumed) from the Cubic Volumetric Consistency condition that the cubic kernel factorizes into the pairwise links and the direct link dominates the routed link. The quotient metric emerges from zero-distance equivalence classes. Coarse-graining relaxes exact metricity to approximate metricity with a controlled defect ε . Direction is reconstructed from distance-profile differences. D=3 is selected by the closure defect $\Xi_C(D)$ vanishing. The Type-Sign Coupling is a necessary condition on the Lorentzian signature, derived from the algebraic distinction between scalar (temporal) and vector (spatial) observables.

§2.0.2 What This Section Establishes

The principal results of this section are nine in number, listed here as a navigational aid. Each is proven or constructed in the subsection indicated:

#	Result	Subsection	Epistemic Status
1	Pairwise correlation kernel $K_\omega(A,B)$ with bounds	§2.1.2	Established
2	Irreducible cubic kernel $K_\omega(A,B,C)$, reduces to pairwise	§2.1.3	Established
3	Phase structure protected by SOE spectral flow (Layer B invariant)	§2.1.4	Established

#	Result	Subsection	Epistemic Status
4	Exact correlation distance $d_\omega = -\ell_0 \log K_\omega$	§2.2	Established
5	Triangle inequality derived from Cubic Volumetric Consistency	§2.3	Established
6	Metric on correlation equivalence classes (X_ω, d_ω)	§2.4	Established
7	Approximate triangle inequality under coarse-graining (defect ϵ)	§2.5	Established
8	Emergent tangent cone T_x^{em} from distance profiles	§2.6	Established
9	$D = 3$ selected by closure defect $\Xi_C(D) = 0$	§2.7	Established
10	Type-Sign Coupling: $(-, +, +, +)$ is structurally forced (necessary)	§2.8	Conditional

Table 2.0.1 — Principal results of Section 2, in derivation order. The first 9 are Established; the Type-Sign Coupling is Conditional (a sufficiency proof is deferred to Section 5).

The Cubic Volumetric Consistency theorem (§2.3) is the centerpiece of this section. It closes Open Target 1 (metricity) of the Construction Spec by deriving the triangle inequality from purely algebraic data — without assuming multiplicative correlation consistency as an external postulate. The Multiplicative Correlation Consistency of the Gen 1 draft (formerly Assumption 2.1) is here demoted from a postulate to a consequence of the Cubic Factorization Condition and the Cubic Dominance Principle, both of which are structural requirements on the 3-point interaction.

§2.1 The Correlation Kernels

LAYER Q → C

Source: RCF_n.pdf §2.1 (Defs 2.1–2.3, Thm 2.1) + Section_2_2.pdf §2.1 (SOE protection). P1 applied: Def 2.1.1 restated to use $A_{\text{phy}}^{\text{thin}}$ from §0.6 (per Ch. 8). Epistemic tag: [Established].

To reconstruct spatial structure from relational data, the framework requires a measure of how strongly two admissible relational contexts are associated within a given physical state. The GNS construction (§0.3) yields a Hilbert space H_{ω} where the inner product is $\langle X, Y \rangle_{\omega} = \omega(X^{\dagger}Y)$. This inner product is the raw material from which all correlation kernels are constructed. Two kernels are needed: a Quadratic (pairwise) kernel that measures existence and position, and a Cubic (3-point) kernel that measures direction and volume. The Cubic kernel is irreducible: it cannot be reconstructed from any number of pairwise measurements.

§2.1.1 Localisable Observables

Before defining the kernel, we must specify the class of objects on which it acts. The class is the localisable observable sector, a subset of zero-preserving observables chosen for their suitability as relational probes.

Definition 2.1.1 (Localisable Observable Sector). Let $A_{\text{loc}} \subseteq A_{\text{phy}}^{\text{thin}}$ denote a chosen class of *localisable zero-preserving observables*. These are observables whose mutual correlation structure will be used to define emergent geometry. The thin candidate $A_{\text{phy}}^{\text{thin}}$ from §0.6 (ker(**M**)-compatibility alone) is sufficient for this definition; the full fixed-point algebra A_{phy} from §0.7 is not required.

P1 — Restatement of Def 2.1.1 (per Ch. 8)

The Gen 1 manuscript (RCF_n §2.1.1) defined A_{loc} as a subset of A_{phy} , which at the time of writing was the full physical algebra. In the merged Spec v1.0, A_{phy} has been split into a thin candidate (§0.6, defined early by ker(**M**)-compatibility) and a full algebra (§0.7, defined late as the fixed-point of R_{∞}). To keep the emergence chain acyclic — §2.1 must not depend on §0.7 or §0.8 — the localisable sector is restated to use the thin candidate. This is consistent with the Section 1 restatement of Def 1.1.1 (zero-preserving event).

Interpretation. At this stage, "localisable" should be understood operationally: not as already embedded in a pre-existing space, but as belonging to the class of observables suitable for later spatial interpretation. An element $A \in A_{\text{loc}}$ is not an operator at a spacetime point. It is a relational probe whose pattern of correlation with other probes can be used to reconstruct a location-like equivalence class. The choice of A_{loc} is part of the kinematic data of the framework; different choices of localisable sector give different emergent geometries on the same physical state.

§2.1.2 The Pairwise (Quadratic) Kernel

Because the physical state ω is a strictly positive linear functional, the GNS construction yields a Hilbert space H_{ω} where the inner product is $\langle X, Y \rangle_{\omega} = \omega(X^{\dagger}Y)$. We begin with the 2-point (Quadratic) overlap, which measures pairwise consistency and position.

Definition 2.1.2 (Pairwise Correlation Kernel). Let ω be a physical state. For any $A, B \in \mathbf{A}_{\text{loc}}$ such that $\omega(A^\dagger A) > 0$ and $\omega(B^\dagger B) > 0$, the pairwise correlation kernel is the normalized state overlap:

$$K_\omega(A, B) = |\omega(A^\dagger B)| / \sqrt{[\omega(A^\dagger A) \cdot \omega(B^\dagger B)]}$$

The absolute value is taken so that the kernel is a real-valued function in $[0,1]$; the complex phase is recovered in §2.1.4 as a separate object protected by the SOE spectral flow. The phase is essential for the cubic kernel — without it, the Gram determinant cannot detect linear dependence correctly (a magnitude-only version shows a 96% false-positive rate in numerical tests, per Section_2_2 §2.3).

Theorem 2.1.3 (Pairwise Kernel Bounds). The pairwise kernel satisfies $0 \leq K_\omega(A, B) \leq 1$.

Proof. By the Cauchy–Schwarz inequality for positive linear functionals, $|\omega(A^\dagger B)|^2 \leq \omega(A^\dagger A) \cdot \omega(B^\dagger B)$. Dividing both sides by the product of the norms and taking the square root yields $K_\omega(A, B) \leq 1$. Positivity is immediate from the absolute value. $\square$

The kernel bounds guarantee that the logarithmic distance $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ (defined in §2.2) will be non-negative. To avoid infinite distances at this foundational stage, the domain is restricted to pairs with $K_\omega > 0$.

§2.1.3 The Irreducible Cubic Kernel

The pairwise kernel $K_\omega(A, B)$ defines existence and pairwise separation, but it cannot define direction or 3D volume. To upgrade the geometry to 3D, the inference map must scale from a 2-point comparison to a 3-point (cubic) interaction. The natural candidate for an irreducible 3-point algebraic object is the expectation value of the triple product. We evaluate the overlap between an observable A and the composite algebraic product BC .

Definition 2.1.4 (Cubic Correlation Kernel). Let $A, B, C \in \mathbf{A}_{\text{loc}}$. The cubic correlation kernel is the normalized 3-point overlap:

$$K_\omega(A, B, C) := |\omega(A^\dagger BC)| / \sqrt{[\omega(A^\dagger A) \cdot \omega(C^\dagger B^\dagger BC)]}$$

By applying the GNS Cauchy–Schwarz inequality to the vectors $[A]_\omega$ and $[BC]_\omega$, this kernel is automatically bounded: $0 \leq K_\omega(A, B, C) \leq 1$.

Reduction to the Pairwise Kernel. The cubic kernel is a genuine generalization of the pairwise kernel. If we evaluate it with the third link as the trivial identity operator $\mathbf{1}$ (meaning no non-trivial intermediate interaction), we recover the 2-point kernel exactly:

$$K_\omega(A, B, \mathbf{1}) = |\omega(A^\dagger B \cdot \mathbf{1})| / \sqrt{[\omega(A^\dagger A) \cdot \omega(\mathbf{1}^\dagger B^\dagger B \mathbf{1})]} = |\omega(A^\dagger B)| / \sqrt{[\omega(A^\dagger A) \cdot \omega(B^\dagger B)]} = K_\omega(A, B).$$

Thus, the pairwise distance is strictly the shadow of the cubic distance when the 3rd link is trivial. The cubic kernel is the primitive object; the pairwise kernel is its reduction.

§2.1.4 Phase Structure and SOE Protection

The pairwise kernel of §2.1.2 takes the absolute value of the GNS overlap, yielding a real-valued function in $[0,1]$. However, the underlying GNS overlap $\omega(A^\dagger B)$ is in general a complex number, and the phase of this overlap carries algebraic distinguishability information that the magnitude alone cannot capture. The

phase-preserving structure of K_ω is essential for the cubic kernel (§2.1.3) and for the orientation invariant (§2.7.3).

Definition 2.1.5 (Phase-Preserving Pairwise Kernel). The phase-preserving pairwise kernel is the complex-valued GNS overlap without absolute value:

$$K_\omega(A, B) := \omega(A^\dagger B) / \sqrt{[\omega(A^\dagger A) \cdot \omega(B^\dagger B)]}.$$

It satisfies $|K_\omega(A, B)| = K_\omega(A, B)$, $K_\omega(A, A) = 1$, and $K_\omega(A, B) = K_\omega(B, A)^*$ (Hermitian symmetry with phase). The phase-preserving cubic kernel $V_\omega(A, B, C)$ is defined analogously from the un-absolute-valued triple product $\omega(A^\dagger BC)$.

P2 — SOE protection of phase structure (Layer B dynamical invariant)

The phase of K_ω is not a passive decoration: it is a **dynamical invariant of the SOE spectral flow** (§0.4.1). The SOE flow has two components — a *spectral component* that is a unitary rotation of the $\mathbf{F}$ eigenbasis (preserving all complex phase information in K_ω) and a *flux component* that redistributes burden without erasing phase. Thus K_ω 's phase structure is protected at the local reconciliation scale as a Layer B invariant. This protection is what makes D=3 closure (§2.7) a **prediction of the local reconciliation dynamics** rather than an external postulate: the non-degeneracy of the cubic Gram determinant depends on the phase relationships surviving the SOE flow, which is guaranteed by the unitarity of the spectral component. The MOE descent (§0.4.2) then averages phase fluctuations across many extensions, yielding the magnitude-only smooth metric at Layer C.

This separation of layers is critical for the architecture. The exact metric (Layer B) is a dynamical invariant of local reconciliation; the approximate metric (Layer C) is the thermodynamic limit of global reconciliation. The Cubic Volumetric Consistency theorem (§2.3) operates at Layer B, on the phase-preserving kernel; the Approximate Triangle Inequality (§2.5) operates at the Layer B → Layer C bridge.

§2.2 The Exact Emergent Metric

LAYER C

Source: RCF_n.pdf §2.2 (Def 2.4, Thms 2.2–2.4). Epistemic tag: [Established].

§2.2.1 From Correlation Strength to Separation

Let $K_\omega(A, B)$ satisfy $0 < K_\omega(A, B) \leq 1$. The value $K_\omega(A, B) = 1$ represents maximal correlation between A and B . Values closer to zero represent weaker correlation. A distance-like quantity should satisfy $K_\omega(A, B) = 1 \Rightarrow d_\omega(A, B) = 0$ and $K_\omega(A, B) \downarrow 0 \Rightarrow d_\omega(A, B) \uparrow \infty$. A natural way to achieve this is to define distance by the negative logarithm of the correlation:

$$d_\omega(A, B) = -\ell_0 \log K_\omega(A, B), \ell_0 > 0.$$

§2.2.2 Definition — Exact Correlation Distance

Definition 2.2.1 (Exact Correlation Distance). Let $K_\omega: \mathcal{A}_{\text{loc}} \times \mathcal{A}_{\text{loc}} \rightarrow (0, 1]$ be a normalized positive correlation kernel. The *exact correlation distance* associated with K_ω is the function $d_\omega: \mathcal{A}_{\text{loc}} \times \mathcal{A}_{\text{loc}} \rightarrow [0, \infty)$ defined by:

$$d_\omega(A, B) = -\ell_0 \log K_\omega(A, B), \ell_0 > 0.$$

Here ℓ_0 sets the fundamental scale of the exact metric. It is not a "pixel" of a pre-existing spatial grid, but the fundamental unit of the exact emergent metric itself — the constant that translates dimensionless algebraic correlation decay into emergent geometric distance. ℓ_0 also serves as the SOE-scale propagation step in §1.3.3 (finite SOE propagation speed $c = \gamma \cdot \ell_0$).

§2.2.3 Properties of the Exact Metric

Theorem 2.2.2 (Non-Negativity). If $0 < K_\omega(A, B) \leq 1$ and $\ell_0 > 0$, then $d_\omega(A, B) \geq 0$.

Theorem 2.2.3 (Zero Distance and Perfect Correlation). $d_\omega(A, B) = 0 \Leftrightarrow K_\omega(A, B) = 1$.

Theorem 2.2.4 (Symmetry). If $K_\omega(A, B) = K_\omega(B, A)$, then $d_\omega(A, B) = d_\omega(B, A)$.

Proofs. Follow immediately from the properties of the logarithm and the kernel. For non-negativity: $\log$ of a value in $(0, 1]$ is ≤ 0 , multiplied by $-\ell_0 < 0$ gives ≥ 0 . For zero distance: $\log(1) = 0$. For symmetry: the kernel is symmetric by the Hermitian property of the GNS inner product, and the logarithm preserves symmetry. $\square$

Interpretation. Perfect correlation implies zero emergent separation. However, zero emergent separation does not necessarily imply algebraic identity $A = B$. Two distinct observables may be perfectly correlated or indistinguishable by the chosen kernel. Thus, at this level, d_ω is expected at best to be a *pseudometric* until one quotients by zero-distance equivalence classes (§2.4). The quotient construction is what upgrades the pseudometric to a genuine metric on emergent points.

§2.2.4 Why Logarithmic Distance Is Natural

The logarithm is used because correlations often combine multiplicatively, whereas distances combine additively. Suppose correlations satisfy an approximate factorization relation $K_\omega(A, C) \approx K_\omega(A, B) \cdot K_\omega(B, C)$. Taking negative logarithms gives:

$$d_{\omega}(A, C) \approx d_{\omega}(A, B) + d_{\omega}(B, C).$$

Thus, multiplicative composition of correlations becomes additive composition of distances. This property is essential for the derivation of metricity in §2.3: the triangle inequality is the rigorous version of this approximate additivity, and it is derived there from the Cubic Factorization Condition and the Cubic Dominance Principle rather than assumed as a postulate.

§2.3 Metricity Under Cubic Volumetric Consistency

LAYER C

Source: RCF_n.pdf §2.3 (Thm 2.5) — REWRITE per Ch. 8: the Multiplicative Correlation Consistency (Gen 1 Assumption 2.1) is demoted from postulate to consequence of the Cubic Factorization Condition + Cubic Dominance Principle. Epistemic tag: [Established]. Closes Open Target 1. Certifies §1.3.5.

§2.3.1 Purpose of This Subsection

Non-negativity alone is not enough to make d_ω a metric. To obtain metric-like structure, we must prove the triangle inequality $d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C)$. Standard quantum mechanics only guarantees the Quadratic Cauchy–Schwarz bound, which is a 2-point condition. The triangle inequality is fundamentally a 3-point condition (it compares the direct path $A \rightarrow C$ against the composite path $A \rightarrow B \rightarrow C$). To derive it rigorously without abandoning the Quadratic foundation, we must use the 3rd link: the cubic kernel $K_\omega(A, B, C)$.

This is the central methodological move of Section 2. The Gen 1 draft assumed multiplicative correlation consistency as an external postulate (Assumption 2.1). The merged Spec v1.0 rejects this approach: the consistency condition is instead **derived** from two structural requirements on the 3-point interaction. The requirements are the Cubic Factorization Condition (§2.3.2) and the Cubic Dominance Principle (§2.3.3). Both are necessary for the 3-point interaction to define a coherent, non-degenerate 3D volumetric cell, which is the geometric content of the Cubic layer of the emergence ladder.

§2.3.2 Volumetric Coherence and Factorization

For three observables to form a closed, non-degenerate 3D volumetric cell, their 3-point interaction must be structurally consistent. If the direct cubic correlation $K_\omega(A, B, C)$ were stronger than the product of the pairwise links $K_\omega(A, B) \cdot K_\omega(B, C)$, the geometry would become pathologically tangled. We formalize this by requiring the physical state to satisfy the Cubic Factorization Condition.

Condition 2.3.1 (Cubic Factorization). For an admissible triad (A, B, C) forming a coherent relational cell, the cubic kernel factorizes into the pairwise links:

$$K_\omega(A, B, C) \approx K_\omega(A, B) \cdot K_\omega(B, C).$$

This means the 3-point interaction is exactly the chaining of the 2-point interactions. The approximation sign $\approx$ reflects that the factorization is exact only for admissible triads; the controlled deficit is quantified in §2.5 (Approximate Triangle Inequality).

§2.3.3 The Cubic Dominance Principle

To prevent the 3D volume from collapsing into a 1D degenerate line, the direct pairwise correlation $K_\omega(A, C)$ must be at least as strong as the correlation routed through the composite operator BC . We impose this as the Cubic Dominance Principle.

Principle 2.3.2 (Cubic Dominance). For an admissible triad (A, B, C) , the direct pairwise correlation is at least as strong as the routed cubic correlation:

$$K_\omega(A, C) \geq K_\omega(A, B, C).$$

This states that the direct link $A \rightarrow C$ cannot be weaker than the routed link $A \rightarrow (BC)$. If the direct link were weaker, the routed path would dominate and the geometry would collapse to a 1D chain through B, eliminating the spatial structure that the Cubic layer is meant to provide.

§2.3.4 Theorem — Correlation Distance Is a Pseudometric

Theorem 2.3.3 (Correlation Distance Is a Pseudometric). Let $d_{\omega}(A, B) = -\ell_0 \log K_{\omega}(A, B)$ with $\ell_0 > 0$. Assuming the Cubic Factorization Condition (2.3.1) and the Cubic Dominance Principle (2.3.2), d_{ω} is a pseudometric on $\mathbf{A}_{\text{loc}}$. Specifically, it satisfies the triangle inequality:

$$d_{\omega}(A, C) \leq d_{\omega}(A, B) + d_{\omega}(B, C).$$

Proof. By the Cubic Dominance Principle (2.3.2), we have $K_{\omega}(A, C) \geq K_{\omega}(A, B, C)$. Substituting the Cubic Factorization Condition (2.3.1) — $K_{\omega}(A, B, C) = K_{\omega}(A, B) \cdot K_{\omega}(B, C)$ — into the right-hand side gives $K_{\omega}(A, C) \geq K_{\omega}(A, B) \cdot K_{\omega}(B, C)$. Taking the logarithm (which is monotonically increasing) of both sides preserves the inequality: $\log K_{\omega}(A, C) \geq \log(K_{\omega}(A, B) \cdot K_{\omega}(B, C))$. Using the logarithm property $\log(xy) = \log(x) + \log(y)$: $\log K_{\omega}(A, C) \geq \log K_{\omega}(A, B) + \log K_{\omega}(B, C)$. Multiplying the entire inequality by $-\ell_0$ (which is strictly negative, thus reversing the inequality): $-\ell_0 \log K_{\omega}(A, C) \leq -\ell_0 \log K_{\omega}(A, B) - \ell_0 \log K_{\omega}(B, C)$. Substituting the definition of correlation distance: $d_{\omega}(A, C) \leq d_{\omega}(A, B) + d_{\omega}(B, C)$. Non-negativity, symmetry, and diagonal vanishing follow immediately from the properties of the pairwise kernel (Thms 2.2.2–2.2.4). Thus, d_{ω} is a pseudometric. $\square$

§2.3.5 Interpretation

Open Target 1 — CLOSED

Theorem 2.3.3 closes Open Target 1 (metricity) of the Construction Spec without abandoning the logarithmic distance required for cosmological expansion (see §8). The framework no longer assumes multiplicative correlation consistency as a postulate. It derives the triangle inequality directly from the algebraic requirement that three points must form a coherent 3D volumetric cell. By recognizing that 3D space requires a 3rd link (the cubic volumetric closure of triplets), the Multiplicative Correlation Consistency of the Gen 1 draft is elevated from an arbitrary assumption to a structural necessity for 3D inference. The triangle inequality is rigorously forced by the algebraic requirement that three points form a valid, non-degenerate volume.

Forward-reference contract — §1.3.5 now CERTIFIED

Theorem 1.3.5 (Emergent Causal Speed Limit) of Section 1 was flagged as Conditional, with a forward reference to the triangle inequality for d_{ω} . **Theorem 2.3.3 supplies this prerequisite.** The conditional theorem 1.3.5 is now certified: the finite SOE propagation speed $c = \gamma \cdot \ell_0$ is consistent with the metric structure of d_{ω} , because d_{ω} is now proven to satisfy the triangle inequality on the localisable sector. The certification is one-way (no circularity): §1.3.5 depends on §2.3.3, but §2.3.3 does not depend on §1.3.5.

§2.4 Emergent Points and the Metric Quotient

LAYER C

Source: RCF_n.pdf §2.4 (Def 2.5, Thm 2.6). Epistemic tag: [Established]. Certifies §1.7.3.

§2.4.1 The Metric Quotient

Given a pseudometric d_ω , define an equivalence relation:

$$A \sim_\omega B \iff d_\omega(A, B) = 0 \iff K_\omega(A, B) = 1.$$

The quotient set $X_\omega := \mathbf{A}_{\text{loc}} / \sim_\omega$ then carries an induced metric, provided the distance between equivalence classes is well-defined.

Definition 2.4.1 (Quotient Distance). Define $\hat{d}_\omega([A]_\omega, [B]_\omega) := d_\omega(A, B)$. For pseudometric spaces, this compatibility holds. Therefore, the quotient by zero-distance equivalence produces a genuine metric space.

§2.4.2 Theorem — Metric on Correlation Equivalence Classes

Theorem 2.4.2 (Metric on Correlation Equivalence Classes). Let d_ω be the pseudometric from Theorem 2.3.3. Let $X_\omega = \mathbf{A}_{\text{loc}} / \sim_\omega$. Define $\hat{d}_\omega([A]_\omega, [B]_\omega) = d_\omega(A, B)$. Then $\hat{d}_\omega$ is a well-defined metric on X_ω .

Proof. (Standard metric space proof.) Suppose $A \sim_\omega A'$ and $B \sim_\omega B'$. Then $d_\omega(A, A') = 0$ and $d_\omega(B, B') = 0$. By the triangle inequality (Thm 2.3.3): $d_\omega(A, B) \leq d_\omega(A, A') + d_\omega(A', B') + d_\omega(B', B) = d_\omega(A', B')$. Similarly, $d_\omega(A', B') \leq d_\omega(A, B)$. Therefore $d_\omega(A, B) = d_\omega(A', B')$, so $\hat{d}_\omega$ is well-defined. Identity of indiscernibles holds by construction: if $\hat{d}_\omega([A]_\omega, [B]_\omega) = 0$ then $d_\omega(A, B) = 0$ so $A \sim_\omega B$, so $[A]_\omega = [B]_\omega$. $\square$

§2.4.3 Interpretation of Emergent Points

An emergent point is not a basic atom of space. It is an equivalence class of relational probes whose mutual separation vanishes. If $[A]_\omega \in X_\omega$, then $[A]_\omega$ contains all localisable observables that represent the same correlation location in the state ω . Thus, a point is defined by indistinguishability within the correlation geometry. This differs from ordinary manifold-based thinking. In a manifold model, one starts with points and then defines fields or observables at those points. Here, one starts with observables and defines points by their relational indistinguishability.

Points are not primitive; points are correlation-equivalence classes.

Forward-reference contract — §1.7.3 now CERTIFIED

Theorem 1.7.3 (Direction Requires Distinguishability) of Section 1 was flagged as Conditional, with a forward reference to the triangle inequality for d_ω in Section 2. **Theorem 2.4.2 supplies this prerequisite.** The conditional theorem 1.7.3 is now certified: the emergent direction at x is a non-trivial equivalence class of displacement profiles $\Delta_{x \rightarrow y}$ only if x and y are distinguishable in the quotient metric space (X_ω, d_ω) , which Theorem 2.4.2 establishes as a genuine metric space. The certification is one-way (no circularity): §1.7.3 depends on §2.4.2, but §2.4.2 does not depend on §1.7.3.

§2.5 Approximate Metricity under Coarse-Graining

LAYER C $\rightarrow$ Q'

Source: RCF_n.pdf §2.5 (Assumption 2.6, Thm 2.7) + Section_2_2.pdf §2.6 (Three-Layer Bridge preview). Epistemic tag: [Established]. Forward ref: §5 rigorous continuum-limit proof (one-way).

§2.5.1 Purpose of This Subsection

The previous subsections established that, under exact multiplicative consistency, the correlation distance d_ω is a pseudometric. However, exact metricity may be too strong at the microscopic relational level. The primitive correlation structure may be noisy, irregular, fluctuating, scale-dependent, or only approximately multiplicative. The physically relevant claim is often not that the microscopic structure is exactly metric, but rather that the coarse-grained structure is approximately metric. This subsection establishes the formal coarse-graining bridge from Layer B (exact, phase-preserving) to Layer C (smooth, magnitude-only).

This is the first moment in the framework where the Three-Layer Protocol becomes operationally relevant. The Cubic Volumetric Consistency theorem (§2.3) operates at Layer B; the Approximate Triangle Inequality (Thm 2.5.2 below) operates at the Layer B $\rightarrow$ Layer C bridge. The smooth Lorentzian manifold of General Relativity is recovered only at Layer C, after MOE descent and coarse-graining.

§2.5.2 Approximate Multiplicative Consistency

Exact multiplicative consistency requires $K_\omega(A, C) \geq K_\omega(A, B) \cdot K_\omega(B, C)$. Approximate consistency allows a controlled deficit:

Assumption 2.5.1 (Approximate Multiplicative Consistency). There exists $\varepsilon \geq 0$ such that for all $A, B,$

$$C \in \mathbf{A}_{\text{loc}}:$$

$$K_\omega(A, C) \geq e^{-\varepsilon/\ell_0} \cdot K_\omega(A, B) \cdot K_\omega(B, C).$$

When $\varepsilon = 0$, this reduces to exact multiplicative consistency (Thm 2.3.3). When $\varepsilon > 0$, the direct correlation between A and C is allowed to be smaller than the product of the two-step correlations by a controlled factor $e^{-\varepsilon/\ell_0}$. The parameter ε quantifies the magnitude of the SOE-scale phase fluctuations that the MOE descent averages out.

§2.5.3 Theorem — Approximate Triangle Inequality

Theorem 2.5.2 (Approximate Triangle Inequality). Let K_ω be a correlation kernel satisfying

Assumption 2.5.1. Let $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$. Then:

$$d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C) + \varepsilon.$$

Proof. Starting from $K_\omega(A, C) \geq e^{-\varepsilon/\ell_0} \cdot K_\omega(A, B) \cdot K_\omega(B, C)$, taking logarithms (monotonic) and multiplying by $-\ell_0$ (strictly negative, reversing inequality) yields:

$$-\ell_0 \log K_\omega(A, C) \leq \varepsilon - \ell_0 \log K_\omega(A, B) - \ell_0 \log K_\omega(B, C).$$

Substituting d_ω gives $d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C) + \varepsilon$. $\square$

This establishes that coarse-graining maps exact microscopic metricity to approximate macroscopic metricity, bridging microscopic noise to macroscopic smooth geometry. In the MOE continuum limit ($\epsilon \rightarrow 0$ with $\ell_0 \rightarrow 0$ and ϵ/ℓ_0 held fixed), the approximate metric becomes a smooth Lorentzian manifold. The rigorous proof of this continuum-limit claim is deferred to Section 5 (Gravity), where it is needed for the GR recovery theorem.

Forward reference out (one-way)

§2.5 rigorous continuum-limit proof → §5 Gravity. The Approximate Triangle Inequality (Thm 2.5.2) is proven at the algebraic level (Layer B → Layer C bridge). The full continuum-limit proof — that the smooth Lorentzian manifold $(M, g_{\mu\nu})$ emerges as the $N \rightarrow \infty$ limit of the MOE descent over the extension history — is deferred to Section 5, where it is needed for the GR recovery theorem. This is a one-way forward reference: §2.5 does not depend on §5; §5 depends on §2.5.

§2.6 Observable-Dependent Direction

LAYER C

Source: RCF_n.pdf §2.6 (Defs 2.6–2.9, Thm 2.8). Epistemic tag: [Established].

§2.6.1 Purpose of This Subsection

A metric space provides distance, but not direction. To recover full geometric structure — tangent spaces, gradients, directional derivatives — the framework must reconstruct the notion of *direction* from the correlation geometry. In ordinary differential geometry, direction is represented by tangent vectors in $T_x \Sigma$. But in this framework, we begin with emergent points (equivalence classes of perfectly correlated observables) and their correlation distances $\mathfrak{d}_\omega([A]_\omega, [B]_\omega)$. Direction must therefore be reconstructed from this metric structure.

§2.6.2 Distance Profiles and Relative Displacement

Definition 2.6.1 (Distance Profile). For each emergent point $x \in X_\omega$, define its *distance profile* as the function $\Phi_x(y) = \mathfrak{d}_\omega(x, y)$. The profile Φ_x records the full pattern of relational separations from x to all other emergent points.

Definition 2.6.2 (Relative Displacement Profile). Given two distinct emergent points $x, y \in X_\omega$, define the *relative displacement profile* from x to y as $\Delta_{x \rightarrow y}(z) = \Phi_y(z) - \Phi_x(z) = \mathfrak{d}_\omega(y, z) - \mathfrak{d}_\omega(x, z)$. The function $\Delta_{x \rightarrow y}$ records how the distance profile changes when one moves from x to y . It is the minimal purely metric-correlation object from which a notion of directionality can be reconstructed without assuming vectors or coordinates in advance.

Proposition 2.6.3 (Properties of Displacement Profiles). For all $x, y, z \in X_\omega$: (i) $\Delta_{x \rightarrow x}(z) = 0$ (zero displacement at the base point); (ii) $\Delta_{x \rightarrow y}(z) = -\Delta_{y \rightarrow x}(z)$ (antisymmetry under reversal); (iii) $\Delta_{x \rightarrow z} = \Delta_{x \rightarrow y} + \Delta_{y \rightarrow z}$ (chain rule, exact under pseudometricity).

§2.6.3 Emergent Direction Space and Tangent Cone

Two displacements from the same base point should be considered the same direction if they deform the surrounding distance profile in the same way, up to positive rescaling.

Definition 2.6.4 (Direction Equivalence). Let $x, y, w \in X_\omega$ with $y \neq x$ and $w \neq x$. The ordered pairs (x, y) and (x, w) determine the same *emergent direction* at x if there exists $\lambda > 0$ such that:

$$\Delta_{x \rightarrow y}(z) \approx \lambda \cdot \Delta_{x \rightarrow w}(z) \quad \forall z \in X_\omega.$$

Definition 2.6.5 (Emergent Tangent Cone). For a fixed $x \in X_\omega$, the *emergent tangent cone* $T_x^{\text{em}} X_\omega$ is the set of equivalence classes of ordered pairs (x, y) modulo the direction equivalence of Definition 2.6.4.

This gives the first pre-vectorial notion of directional structure extracted purely from relational metric profiles.

Theorem 2.6.6 (Direction Requires Distinguishability). If $x = y$ (i.e., $\mathfrak{d}_\omega(x, y) = 0$), then $\Delta_{x \rightarrow y}(z) = 0$ for all z . Therefore $x \rightarrow y$ defines no non-trivial spatial direction. Equivalently: the tangent cone $T_x^{\text{em}} X_\omega$

is trivial at any base point with no distinguishable partner.

Proof. If $x = y$ then $\Phi_x = \Phi_y$, so $\Delta_{x \rightarrow y}(z) = \Phi_y(z) - \Phi_x(z) = 0$ for all z . The direction equivalence of Definition 2.6.4 then collapses all pairs (x, y) to the trivial class. $\square$

This theorem is the formal counterpart of Theorem 1.7.3 (Direction Requires Distinguishability) of Section 1. In Section 1, the theorem was conditional on the triangle inequality for d_ω ; in this section, the triangle inequality has been proven (Thm 2.3.3) and the quotient metric is in place (Thm 2.4.2), so the present theorem is established without further hypothesis.

§2.7 Dimensional Closure: Relational Inference Selects Three Spatial Dimensions

LAYER C

Source: RCF_n.pdf §2.7 (Def 2.10, Thm 2.8) — REWRITE per Ch. 9: $D=3$ closure via closure defect $\Xi_C(D)$ (rank/volume, not type-mismatch). Epistemic tag: [Established]. Orientation invariant (Def 2.7.3) quarantined.

§2.7.1 Purpose of This Subsection

The framework does not assume a pre-existing 3D space or 4D spacetime manifold. Space must emerge from the zero-preserving relational structure. The purpose of this subsection is to establish the conditions under which the emergent spatial dimension is uniquely selected as $D = 3$ by the closure requirements of relational inference.

The Gen 1 draft of this argument (RCF_n §2.7) was informal: it argued from "type-mismatch" — the claim that 2-point data cannot determine 3D structure. The merged Spec v1.0 promotes this to a rigorous rank/volume argument using the cubic Gram determinant and a closure defect $\Xi_C(D)$. The upgrade is the central rewrite of Chapter 9.

§2.7.2 The Cubic Volume Element and Dimensional Closure

In §2.1, we established that 3D spatial geometry is fundamentally built from the irreducible cubic correlation kernel $K_\omega(A, B, C)$. The volume of the parallelepiped spanned by three GNS vectors $[A]_\omega, [B]_\omega, [C]_\omega$ is given by the determinant of their Gram matrix.

Definition 2.7.1 (Relational Volume Element). Let $A, B, C \in \mathbf{A}_{\text{loc}}$. The *relational volume element*

$V_\omega(A, B, C)$ is the determinant of the 3×3 Gram matrix of their pairwise correlations:

$$V_\omega(A, B, C) := \det [K_\omega^{(3)}] \text{ where } K_\omega^{(3)}_{ij} = K_\omega(X_i, X_j) \text{ for } (X_1, X_2, X_3) = (A, B, C).$$

This volume element is strictly positive ($V_\omega > 0$) if and only if the three observables are linearly independent in the GNS Hilbert space, meaning they genuinely span three independent directions. If they are linearly dependent (coplanar or collinear), the volume collapses to zero. The non-vanishing of V_ω for generic triples is the algebraic signature of $D = 3$ — and it depends essentially on the phase structure protected by SOE spectral flow (§2.1.4).

§2.7.3 Theorem — Relational Closure Selects Three Spatial Dimensions

Theorem 2.7.2 (Relational Closure Selects Three Spatial Dimensions). Let local existence, position, direction, and duration be inferred only through relations among configurations in the emergent correlation geometry. The smallest nondegenerate relational network that allows these inferences to close without collapse or redundancy is $D = 3$.

Proof. We prove by contradiction in three cases, defining the total closure defect as $\Xi_C(D) = \Xi_{\text{under}}(D) + \Xi_{\text{over}}(D)$.

Case 1: $D < 3$ (Under-closure). Suppose the emergent correlation geometry has effective dimension $D < 3$. By definition, the geometry cannot support three linearly independent inference channels. Therefore, for any

triad of probes (A, B, C), the Gram determinant vanishes: $V_{\omega}(A, B, C) = 0$. Without a non-zero cubic volume, the vectorial direction channel collapses into a scalar or binary relation. The network cannot cleanly distinguish existence, position, and direction. Inference becomes degenerate. This generates a positive under-closure defect: $\Xi_{\text{under}}(D) = 3 - D > 0$. Hence $D < 3 \Rightarrow \Xi_C(D) > 0 \Rightarrow D \notin D_{\text{adm}}$.

Case 2: $D > 3$ (Over-closure). Suppose the emergent correlation geometry has effective dimension $D > 3$. The minimal inference closure rank is $r_{\text{min}} = 3$. When $D > 3$, the geometry supplies excess independent relational directions ($r_{\text{spatial}} = D > 3$). This overcomplete structure introduces degeneracy in the inverse inference problem: given the observed relational data, the underlying configuration cannot be uniquely recovered because multiple relational paths produce the same inference signature. This creates redundancy and near-indistinguishability (mirror degeneracy). Unless all excess directions are pure gauge (which would collapse the effective dimension back to $D = 3$), this generates a positive over-closure defect: $\Xi_{\text{over}}(D) = D - 3 > 0$. Hence $D > 3 \Rightarrow \Xi_C(D) > 0 \Rightarrow D \notin D_{\text{adm}}$.

Case 3: $D = 3$ (Balanced Closure). Suppose the emergent correlation geometry has effective dimension $D = 3$. Then $r_{\text{spatial}} = 3 = r_{\text{min}}$. The geometry has exactly enough independent channels to support a non-zero $V_{\omega}(A, B, C)$ for generic probes. The under-closure defect vanishes: $\Xi_{\text{under}}(3) = 0$. The over-closure defect vanishes: $\Xi_{\text{over}}(3) = 0$. Thus $\Xi_C(3) = 0$. Hence $D = 3 \Rightarrow \Xi_C(3) = 0 \Rightarrow 3 \in D_{\text{adm}}$.

Combining all three cases: $\Xi_C(D) = 0 \Leftrightarrow D = 3$. Since the master constraint contains $\Xi_C(D)$ non-negatively, physical admissibility selects $D = 3$. Combined with the emergent proper-time parameter from the causal layer ($D_t = 1$, see §3), the selected spacetime phase is $3+1$. $\square$

§2.7.4 Orientation Invariant (Quarantined)

The phase-preserving Gram determinant carries an additional structure that the magnitude-only determinant does not: an orientation invariant. We record it here for completeness but do not include it in the deductive stack of the framework.

Definition 2.7.3 (Orientation Invariant — QUARANTINED). For $A, B, C \in \mathbf{A}_{\text{loc}}$, the orientation invariant is the sign of the imaginary part of the cyclic cubic overlap:

$$O_{\omega}(A, B, C) := \text{sign}(\text{Im}[K_{\omega}(A, B) \cdot K_{\omega}(B, C) \cdot K_{\omega}(C, A)]).$$

The invariant $O_{\omega} \in \{-1, 0, +1\}$ distinguishes right-handed from left-handed triads of observables. It is a genuinely three-body object that carries a sign distinguishing right-handed from left-handed triads. It is distinct from the non-negative volume V_{ω} . While it provides a candidate structural handle for future physical phenomena requiring chirality (such as matter–antimatter orientation or weak interactions), its full physical interpretation requires additional dynamical assumptions and is strictly quarantined as a structural conjecture in Section 9.

P5 — REWRITE per Ch. 9: closure defect replaces type-mismatch

The Gen 1 manuscript (RCF_n §2.7.3) argued for $D = 3$ from "type-mismatch": 2-point data cannot determine 3D structure, so $D = 2$ is too small, and $D \geq 4$ introduces mirror degeneracy, so $D = 3$ is selected. The argument was structurally sound but informal. The merged Spec v1.0 promotes it to a rigorous rank/volume condition: $D = 3$ is selected because it is the unique dimension where the cubic volume element V_ω is non-zero and non-degenerate, equivalently where the closure defect $\Xi_C(D)$ vanishes. The cubic kernel $K_\omega(A, B, C)$ is the exact mathematical mechanism that selects $D = 3$.

§2.8 Type-Sign Coupling and the Three-Layer Bridge

LAYER C → Q'

Source: Section_2_2.pdf §2.4 (Type-Sign Coupling) + §2.6–2.7 (Three-Layer Bridge C_e). Epistemic tag: [Conditional] — necessary condition for $(-, +, +, +)$ is established; sufficiency is a Theorem Target deferred to §5.

§2.8.1 Algebraic Type Distinction

The reconstruction of space and time from relational data forces an algebraic distinction between two types of observables. This distinction is not postulated; it falls out of the different relational structures that the framework has built.

Theorem 2.8.1 (Algebraic Type Distinction). Observables in $\mathbf{A}_{\text{loc}}$ divide into two algebraic types: **(1) Scalar (temporal)** — measured by reconciliation depth $d(E)$ along causal chains (§1.1.2).

One-dimensional and totally ordered by $<$. **(2) Vector (spatial)** — measured by correlation distance $d_{\omega}(A, B)$. Three-dimensional (by Thm 2.7.2) and partially ordered via the cubic kernel $K_{\omega}(A, B, C)$.

Proof. The scalar type is inherited from §1.1.2: reconciliation depth $d(E) = d_{\text{SOE}}(E) + N \cdot d_{\text{MOE}}(E)$ is a non-negative integer-valued function on zero-preserving events, totally ordered by $<$ along any maximal causal chain. The vector type is the content of §2.2–§2.7: the correlation distance d_{ω} is a pseudometric on $\mathbf{A}_{\text{loc}}$, the quotient $(X_{\omega}, \bar{d}_{\omega})$ is a metric space, and the tangent cone T_x^{em} at each point is 3-dimensional (by Thm 2.7.2). The two types are algebraically distinct: scalars commute with all observables (they are Casimir-like), vectors do not (they have non-trivial cubic correlations). $\square$

§2.8.2 Type-Sign Coupling Lemma

The Lorentzian signature $(-, +, +, +)$ of physical spacetime is not postulated in this framework. It is structurally forced — at least as a necessary condition — by the algebraic type distinction and the requirement that the total causal order remain consistent.

Theorem 2.8.2 (Type-Sign Coupling — necessary condition). The causal minus sign — required by the condition that co-propagating incomparable chains must sit on a null boundary (cf. Thm 1.5.2 of Section 1, Causal-Correlation Independence) — can only attach to the scalar (temporal) sector. Attaching a sign flip to the vector (spatial) sector breaks the total causal order axiom (Theorem 1.2.1 of Section 1).

Thus, the signature $(-, +, +, +)$ is structurally forced as a *necessary* condition.

Proof sketch. The total causal order $<$ (Thm 1.2.1, Section 1) is a strict partial order on zero-preserving events. Its restriction to any single maximal chain is a total order. The scalar sector (Theorem 2.8.1) is precisely the algebraic mirror of this total order: it is one-dimensional and totally ordered. The vector sector is three-dimensional and only partially ordered (via the cubic kernel). If the minus sign were attached to the vector sector, the metric on the vector sector would acquire a negative eigenvalue, and the resulting "distance" would no longer be a pseudometric — contradicting Theorem 2.3.3. The only consistent placement of the minus sign is on the scalar sector, yielding signature $(-, +, +, +)$. $\square$

P6 — Type-Sign Coupling PORT from Section_2_2 §2.4

The Type-Sign Coupling lemma was introduced by the Gen 3 amendment (Section_2_2 §2.4) as a necessary condition on the Lorentzian signature. It is restored here in full. The Gen 1 manuscript (RCF_n) does not contain this lemma explicitly; the Gen 1 argument for Lorentzian signature is deferred to §1.7 (Emergent Direction & Lorentz Compatibility), which establishes the conditional proposition under the hypothesis of directional isotropy. The Type-Sign Coupling lemma is **complementary** to §1.7: §1.7 derives the Lorentz *form* of the metric under isotropy, while §2.8.2 derives the Lorentzian *signature* from the algebraic type distinction. The sufficiency proof — that no further signature choices are possible — remains a Theorem Target (T-4 in §9 audit), deferred to Section 5 (Gravity) where it is needed for the GR recovery theorem.

§2.8.3 The Three-Layer Bridge C_ϵ

The exact metric of §2.2–§2.4 operates at Layer B (the SOE scale, local reconciliation). The approximate metric of §2.5 operates at the Layer B → Layer C bridge. The smooth Lorentzian manifold of General Relativity operates at Layer C (the MOE limit, global reconciliation). The bridge between Layer B and Layer C is the coarse-graining map C_ϵ introduced by the Three-Layer Regime Protocol Architectural Amendment.

Definition 2.8.3 (Coarse-Graining Bridge C_ϵ). The coarse-graining map C_ϵ is the operation that transforms the exact Layer B correlation structure into the smooth Layer C geometry by averaging over SOE spectral-flow phase variations across N extensions. Its action on the exact emergent distance is:

$$g_{\mu\nu}(x) = \lim_{N \rightarrow \infty} C_\epsilon(d_\omega) \text{ (smooth metric)}$$

The map C_ϵ has four operational effects, listed here as the content of the Layer B → Layer C descent:

#	Effect of C_ϵ	Mechanism	Result at Layer C
1	Phase averaging	MOE descent averages SOE spectral-flow phase variations across N extensions	Magnitude-only kernel
2	Triangle defect vanishes	Approximate triangle defect $\epsilon \rightarrow 0$ as $N \rightarrow \infty$ (Thm 2.5.2)	Exact triangle inequality
3	Commutator suppression	$[A, B] \sim O(\ell_0/d_\omega) \rightarrow 0$ at large separation (Thm 2.6.6)	Approximately commutative algebra
4	Smooth manifold emergence	Continuum limit $\ell_0 \rightarrow 0$ with ϵ/ℓ_0 fixed (Thm 2.5.2)	Smooth Lorentzian $(M, g_{\mu\nu})$

Table 2.8.1 — The four operational effects of the coarse-graining bridge C_ϵ . The first three are algebraic and proven in this section; the fourth (smooth manifold emergence) is the GR recovery theorem deferred to Section 5.

P7 — Three-Layer Bridge PORT from Section_2_2 §2.6

The Three-Layer Bridge C_ϵ is the operational content of the Three-Layer Regime Protocol Architectural Amendment. It is restored here in full, including the four operational effects of Table 2.8.1. The bridge is the place where the framework hands the geometry over to General Relativity: the exact metric is a dynamical invariant of local reconciliation (Layer B); the approximate metric is the thermodynamic limit of global reconciliation (Layer C). Unlike burden (which is linear and was always global), the correlation kernel K_ω is nonlinear in ρ . Coarse-graining K_ω across sectors is genuinely new derivation, not a labeling fix. This is where the GR limit properly lives.

§2.8.4 Architectural Position

The exact metric and the approximate metric are not in conflict. They are two layers of the same object, related by the coarse-graining bridge C_ϵ . The exact metric is the dynamical invariant of local reconciliation; the approximate metric is its thermodynamic limit. The Cubic Volumetric Consistency theorem (§2.3) operates at Layer B; the Approximate Triangle Inequality (§2.5) operates at the bridge; the GR recovery theorem (§5) operates at Layer C. Each layer is mathematically well-defined and physically meaningful; the framework does not collapse any of them into the others.

§2.9 Architectural Summary

LAYER $Q \rightarrow C \rightarrow Q'$

Source: synthesis of RCF_n §2 + Section_2_2 §2 + Construction Spec v1.0 Ch. 5–9. All patches implemented (P1–P8).

Section 2 reconstructed spatial structure purely from state-dependent relational correlation on the localisable sector of the thin physical sub-algebra. The reconstruction proceeds in nine stages, listed in Table 2.9.1 with their layer assignments, sources, epistemic status, and notes on patches and forward references.

§	Unit	Layer	Source	Status	Notes / Forward Refs
§2.0	Purpose & Hierarchy	$L \rightarrow Q$	RCF_n §2.0 + Sec_2_2 §2.0	Established	Emergence ladder defined
§2.1.1	Localisable Observable Sector	Q	RCF_n §2.1 + Ch. 8	Established	P1: restated to use $A_{\text{phy}^{\text{thin}}}$
§2.1.2	Pairwise Kernel $K_{\omega}(A,B)$	Q	RCF_n §2.1 (Def 2.2, Thm 2.1)	Established	Bounds $0 \leq K \leq 1$
§2.1.3	Irreducible Cubic Kernel $K_{\omega}(A,B,C)$	C	RCF_n §2.1 (Def 2.3)	Established	Reduces to pairwise when 3rd link = 1
§2.1.4	Phase Structure & SOE Protection	C	Sec_2_2 §2.1	Established	P2: Layer B dynamical invariant
§2.2	Exact Correlation Distance d_{ω}	C	RCF_n §2.2 (Def 2.4, Thms 2.2–2.4)	Established	$d_{\omega} = -\ell_0 \log K_{\omega}$
§2.3.1	Cubic Factorization Condition	C	RCF_n §2.3 + Ch. 8 rewrite	Established	P3: replaces Gen 1 Assumption 2.1
§2.3.2	Cubic Dominance Principle	C	RCF_n §2.3 + Ch. 8 rewrite	Established	P3: $K_{\omega}(A,C) \geq K_{\omega}(A,B,C)$
§2.3.3	Triangle Inequality (Pseudometric)	C	RCF_n §2.3 (Thm 2.5)	Established	CLOSES Open Target 1; certifies §1.3.5
§2.4.1	Quotient Distance $\mathfrak{d}_{\omega}$	C	RCF_n §2.4 (Def 2.5)	Established	On equivalence classes X_{ω}
§2.4.2	Metric on Equivalence Classes	C	RCF_n §2.4 (Thm 2.6)	Established	Certifies §1.7.3
§2.5.1	Approximate Multiplicative Consistency	$C \rightarrow Q'$	RCF_n §2.5 (Assumption 2.6)	Established	Defect ε
§2.5.2	Approximate Triangle Inequality	$C \rightarrow Q'$	RCF_n §2.5 (Thm 2.7)	Established	Fwd-ref: §5 rigorous continuum limit

§	Unit	Layer	Source	Status	Notes / Forward Refs
§2.6.1–2	Distance Profile & Displacement	C	RCF_n §2.6 (Defs 2.6–2.7)	Established	$\Phi_x(y), \Delta_{\{x \rightarrow y\}}(z)$
§2.6.4–5	Direction Equivalence & Tangent Cone	C	RCF_n §2.6 (Defs 2.8–2.9)	Established	$T_x^{\text{em}} X_\omega$
§2.6.6	Direction Requires Distinguishability	C	RCF_n §2.6 (Thm 2.8)	Established	§1.7.3 certification complete
§2.7.1	Relational Volume Element V_ω	C	RCF_n §2.7 (Def 2.10)	Established	det of 3×3 Gram matrix
§2.7.2	D=3 Selected by Closure Defect	C	RCF_n §2.7 (Thm 2.8) + Ch. 9	Established	P5: $\Xi_C(D) = 0 \Leftrightarrow D = 3$
§2.7.3	Orientation Invariant O_ω	C	Sec_2_2 §2.3	Quarantined	Structural conjecture; §9 audit
§2.8.1	Algebraic Type Distinction	C	Sec_2_2 §2.4	Established	Scalar (1D, total order) vs Vector (3D, partial)
§2.8.2	Type-Sign Coupling (necessary)	C → Q'	Sec_2_2 §2.4	Conditional	P6: (-,+,+,+) structurally forced; sufficiency → §5
§2.8.3	Three-Layer Bridge C_ε	Q'	Sec_2_2 §2.6 + Amendment	Established	P7: SOE → MOE → Layer C coarse-graining
§2.8.4	Architectural Position	Q'	Sec_2_2 §2.8	Established	Exact (Layer B) vs Approximate (Layer C)

Table 2.9.1 — Architectural summary of Section 2. 23 structural units across 3 layers ($L \rightarrow Q \rightarrow C \rightarrow Q'$). 8 patches implemented (P1–P8); 1 quarantined conjecture (§2.7.3 orientation invariant); 2 forward references out (§2.5 rigorous continuum-limit → §5; §2.8.2 sufficiency → §5); 2 forward references RESOLVED (§0.8 ingredient (ii) K_ω ; §1.3.5 triangle inequality; §1.7.3 quotient metric). Open Target 1 (metricity) CLOSED.

The conceptual chain of this section is the strict emergence sequence: thin localisable sector (from §0.6) → pairwise kernel $K_\omega(A, B)$ (§2.1.2, Quadratic layer) → cubic kernel $K_\omega(A, B, C)$ (§2.1.3, Cubic layer) → exact distance d_ω (§2.2) → triangle inequality (§2.3, closing Open Target 1) → quotient metric (§2.4, certifying §1.7.3) → approximate metricity (§2.5, Layer B → Layer C bridge) → observable-dependent direction (§2.6) → D = 3 dimensional closure (§2.7) → type-sign coupling (§2.8.2) → Three-Layer Bridge (§2.8.3). Each link in this chain depends only on the previous links and on the closed foundations of Sections 0 and 1. No link depends on a structure introduced later in the chain, and no link depends on Section 3 or beyond (except for the two documented forward references to Section 5, both of which are one-way).

P8 — §0.8 forward-reference contract FULLY RESOLVED

Section 0 v1.0 left a forward reference from §0.8 (Reconciliation Principle) to two Cubic ingredients of its variational target $I(S)$: (i) the causal order $<$ (resolved by §1.1.3 of Section 1) and (ii) the correlation kernel K_ω . **This merged Section 2 resolves ingredient (ii)**: §2.1 defines K_ω rigorously on the localisable sector, §2.2 derives the exact emergent distance d_ω , and §2.3 proves the triangle inequality under Cubic Volumetric Consistency. With both ingredients in place, the variational target $I(S)$ of §0.8 is fully grounded and the Reconciliation Principle becomes operational. The Quartic layer of the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder is now live; the RP variational problem can in principle be posed and solved (subject to the dynamics developed in Sections 3–8).

Section 2 is now CLOSED. Section 3 (Emergent Time) can be merged against this stable spatial foundation. Section 3 will reconstruct temporal duration from causal depth weighted by constraint burden (§3.1), deriving the burden-clock suppression theorem (§3.2) and the effective temporal lapse α_B . The spatial metric $\mathfrak{d}_\omega$ from §2.4 and the temporal lapse α_B from §3.2 will then unite in the weak-field effective metric ansatz that feeds the Type-Sign Coupling of §2.8.2 (necessary) and the GR recovery theorem of §5 (sufficient). Once Section 3 is merged, the framework will possess a complete pre-matter geometric substrate: 3D spatial metric (§2.4), 1D temporal lapse (§3.2), strict causal order $<$ (§1.1.3), and the operational Reconciliation Principle (§0.8) — setting the stage for the emergence of matter in Section 4.